

BFS OUTPUT :

```
● @20525pcs006-del →/workspaces/pythonn (main) $ python bfs.py
Route found:
central -> park -> Airport
○ @20525pcs006-del →/workspaces/pythonn (main) $ █
```

DFS OUTPUT :

```
● @20525pcs006-del →/workspaces/pythonn (main) $ python dfs.py
Main Menu
Withdraw
withdrawal
Savings
Current
Balance Check
PIN Change
○ @20525pcs006-del →/workspaces/pythonn (main) $ █
```

A*OUTPUT:

```
● @20525pcs006-del →/workspaces/pythonn (main) $ python A*.py

Random students:
[Student with 42 marks., Student with 36 marks., Student with 78 marks., Student
with 76 marks., Student with 50 marks.]
Marks of first student:
42

Top student: Student with 78 marks.
-
```

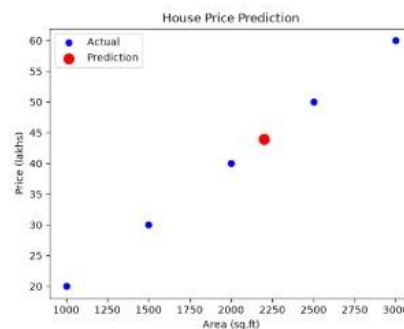
NAÏVE BAYES OUTPUT:

```
● @20525pcs006-del →/workspaces/pythonn (main) $ python naive_bayes
Prediction : Don't play cricket
○ @20525pcs006-del →/workspaces/pythonn (main) $ █
```

BAYESIAN NETWORK OUTPUT :

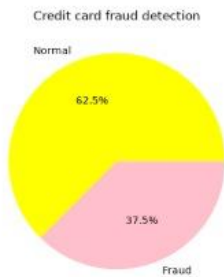
```
● @20525pcs006-del →/workspaces/pythonn (main) $ python bayesian_network
Car Engine Analysis
-----
P(Engine fault) = 0.25
P(Noise | Engine fault) = 0.85
P(Noise) = 0.4
Probability of Engine Fault:
0.53
█
```

REGRESSION MODEL OUTPUT:



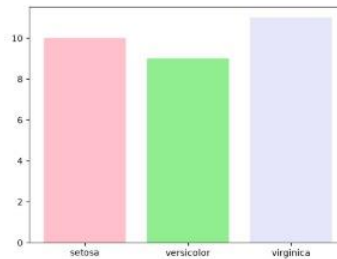
```
Problems  Output  Debug Console  Terminal  Ports
@20525pcs006-del →/workspaces/pythonn (main) $ python regression.py
Enter Area: 2200
Price: ₹ 4400000.0
```

DECISION TREE OUTPUT:



```
Problems Output Debug Console Terminal Ports
@20525pcs006-del →/workspaces/pythonn (main) $ python decision_tree
Fraud Transaction
```

RANDOM FOREST OUTPUT:



```
Problems Output Debug Console Terminal Ports
@20525pcs006-del →/workspaces/pythonn (main) $ python random_forest
Accuracy: 1.0
-setosa- 10
-versicolor- 9
-virginica- 11
```

SVM MODEL OUTPUT:

```
@20525pcs006-del →/workspaces/pythonn (main) $ python svm.py
Enter humidity: 68
Enter pressure: 1009
Enter wind speed: 14
Enter previous temperature: 30
Predicted temperature: 31.53
@20525pcs006-del →/workspaces/pythonn (main) $
```

ENSEMBLING LEARNING OUTPUT:

```
● @20525pcs006-del →/workspaces/pythonn (main) $ python ensemble_learning
----- Battery Health Prediction -----
Accuracy: 100.0 %

Usage Hours: 3
Charging Hours: 1
Temperature: 38
Battery Cycles: 100

Prediction: Healthy
- - - - -
```

CLUSTERING ALGORITHM OUTPUT:

```
● @20525pcs006-del →/workspaces/pythonn (main) $ python clustering_algorithm
----- Enter Smartphone Usage Details -----
Daily Usage Hours : 9
Number of Apps    : 10
Battery Usage (%) : 98

----- Smartphone Usage Prediction -----
Daily Usage Hours : 9.0
Number of Apps    : 10
Battery Usage     : 98.0 %
Usage Category    : HIGH USAGE
```

EM BAYESIAN OUTPUT:

```
● @20525pcs006-del →/workspaces/pythonn (main) $ python em_bayesian.py
Enter CGPA: 8.2
Enter Skill Score: 85
Enter Interview Score: 88

===== EM BAYESIAN NETWORK =====

## EM Analysis
Cluster 0 Probability : 1.0
Cluster 1 Probability : 0.0
Most Likely Cluster   : 0

## Bayesian Prediction
Selection Probability : 100.0 %
Rejection Probability : 0.0 %

## Decision
Candidate Status      : SELECTED
Reason : Good CGPA, Skill and Interview Score

Prediction Completed Successfully
```

NN MODEL OUTPUT:

```
● @20525pcs006-del →/workspaces/pythonn (main) $ python nn_model
Enter Refrigerator Temperature: 6
Enter Door Opening Count: 12
Enter Usage Hours: 8
Enter Outside Temperature: 28
Predicted Energy Consumption: MEDIUM
```

DEEP LEARNING NN MODEL OUTPUT:

```
● @20525pcs006-del →/workspaces/pythonn (main) $ python deeplearning_nn
Enter Days: 30
Enter Temperature: 27
Enter Humidity: 45

----- Cosmetic Product Risk Prediction -----
Opened Days : 30
Temperature : 27 °C
Humidity    : 45 %
Risk Level  : RISK
```